

Why synchronous averaging removes $s(t)$: the $1/\sqrt{N_r}$ leakage, and what heartbeat/breath-driven EEG does to it

Working note for the CFA ear-EEG paper

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Two things up front. (i) The symbol in “ $1/\sqrt{N_r}$ ” is the square-root radical $\sqrt{\cdot}$, not a variable “ v ”. It means *one over the square root of the number of averaged beats N_r* . (ii) Variance does appear, but underneath the radical: the leftover neural/noise in the template has *standard deviation* $SD/\sqrt{N_r}$, where $SD = \sqrt{\text{Var}}$ is the standard deviation of the R-asynchronous part of the signal. The mechanism is *phase-asynchrony*, not sampling rate — the “1 Hz” intuition predicts the wrong thing for alpha and is corrected below.

1 The estimator is a sample mean at fixed phase

For a fixed phase index p (offset from the R-peak), the template value is the average, across N_r beats, of the EEG sampled at the *same* offset p from each detected R-peak $\hat{t}_k$:

$$\hat{d}_r(p) = \frac{1}{N_r} \sum_{k=1}^{N_r} y_k(p), \quad y_k(p) \equiv y(\hat{t}_k + p/f_s). \quad (1)$$

Substituting the signal model $y = d + s + n$ and writing $s_k(p) = s(\hat{t}_k + p/f_s)$, $n_k(p) = n(\hat{t}_k + p/f_s)$:

$$\hat{d}_r(p) = \underbrace{d(p)}_{\text{same every beat}} + \frac{1}{N_r} \sum_{k=1}^{N_r} s_k(p) + \frac{1}{N_r} \sum_{k=1}^{N_r} n_k(p). \quad (2)$$

Note this is *not* decimation to 1 Hz. As p ranges over the window, the template is reconstructed at full rate f_s ; only within a fixed p is there one sample per beat. So the question is never “what is s at 1 Hz” — it is whether the N_r samples $\{s_k(p)\}_k$ at that offset add coherently or cancel.

2 Bias: what survives averaging

Take the expectation of Eq. (2) over beats. The disturbance term is deterministic and R-locked, so it passes through untouched. The neural/noise terms survive *only if they have nonzero mean at the fixed R-phase*:

$$\mathbb{E}[\hat{d}_r(p)] = d(p) + \mathbb{E}[s_k(p)] + \mathbb{E}[n_k(p)]. \quad (3)$$

This is the pivotal line. Whether a component survives is decided by *phase-coherence with the R-peak*, not by its frequency:

- **R-asynchronous** ($\mathbb{E}[s_k(p)] = 0$): across beats the cardiac phase relative to the ongoing EEG rhythm is uniformly distributed, so s is sampled at scattered phases and averages to zero in the mean. *Removed.* (Free-running alpha, background, sensor noise.)

- **R-coherent** ($\mathbb{E}[s_k(p)] \neq 0$): a component time-locked to the heartbeat adds coherently and *is retained in the template* — even if it is neural. (The heartbeat-evoked potential; see §6.)

Why alpha dies although $10\text{ Hz} \gg 1\text{ Hz}$ (the corrected intuition). Let the alpha phase at successive R-peaks be θ_k . Between beats it advances by $\theta_{k+1} - \theta_k = 2\pi f_\alpha \text{RR}_k \bmod 2\pi$. Because $f_\alpha \text{RR}$ is not a low-order rational and both f_α and RR drift, this is an ergodic walk that fills $[0, 2\pi)$ uniformly. At fixed offset p you therefore sample alpha at effectively uniform random phases, whose mean is zero. **Alpha is removed because it is phase-incoherent with the R-peak, not because it is “fast”.** A component’s spectral content is irrelevant; its *relationship to the index clock* is everything. This is exactly RRO/NRRO separation in disk servo: repeatable runout (index-locked) survives synchronous averaging; non-repeatable runout averages out regardless of its frequency band.

3 Variance: where the $1/\sqrt{N_r}$ comes from

The mean removes R-asynchronous components in expectation, but on a *finite* record a residual survives. Its size is a variance calculation. Let the asynchronous part at offset p have per-sample variance

$$\text{Var}[s_k(p) + n_k(p)] = V(p), \quad \text{SD}(p) = \sqrt{V(p)}. \quad (4)$$

The leakage in the template is the sample-mean of these N_r terms. If the per-beat samples are *mutually uncorrelated*,

$$\text{Var}[\hat{d}_r(p)] = \frac{1}{N_r^2} \sum_{k=1}^{N_r} \text{Var}[s_k(p) + n_k(p)] = \frac{V(p)}{N_r}, \quad \boxed{\text{SD}[\hat{d}_r(p)] = \frac{\text{SD}(p)}{\sqrt{N_r}}} \quad (5)$$

That is the $1/\sqrt{N_r}$: the *standard deviation* of the surviving neural/noise leakage shrinks as one over the square root of the beat count, because averaging N_r uncorrelated terms divides their variance by N_r and hence their SD by $\sqrt{N_r}$. Variance sits inside the radical; the radical is why it is $\sqrt{N_r}$ and not N_r .

Numerically: with $N_r = 717$ beats, ideal leakage is $1/\sqrt{717} \approx 3.7\%$ of the single-beat background SD. Your template deflections are an order of magnitude above that floor, which is *why* the R-locked deflection is visible at all — the averaging has bought you $\sim 27\times$ SNR on the disturbance relative to the asynchronous background.

The realistic version: correlated beats and N_{eff}

Equation (5) assumed uncorrelated beats. Real EEG has weak beat-to-beat correlation (next section), so the general result carries a correlation-inflation factor:

$$\text{Var}[\hat{d}_r(p)] = \frac{V(p)}{N_r} \left(1 + \frac{2}{N_r} \sum_{i < j} c_{ij}\right) \equiv \frac{V(p)}{N_{\text{eff}}}, \quad N_{\text{eff}} = \frac{N_r}{1 + \frac{2}{N_r} \sum_{i < j} c_{ij}}, \quad (6)$$

where c_{ij} is the correlation of the asynchronous part between beats i and j . Positive correlation (things that persist across neighboring beats) makes $N_{\text{eff}} < N_r$, so the true leakage is $\text{SD}/\sqrt{N_{\text{eff}}} > \text{SD}/\sqrt{N_r}$. This is the gap between the ideal 3.7% and the empirical $\sim 10\%$ alpha leakage you observe: you do not have 717 *independent* looks, you have fewer, because respiration correlates neighboring beats.

4 How heartbeat- and breath-driven *real* EEG is affected

Not everything physiological is R-asynchronous. Two mechanisms couple genuine EEG to the cardiac and respiratory cycles, and they hit the template in two *different* ways — one as bias, one as variance. This is the distinction the “1 Hz” picture cannot see.

(a) Heartbeat-evoked potential (HEP) — a *bias* term

The HEP is a cortical response *triggered by* each heartbeat (baroreceptor / interoceptive pathways), so it is genuinely time-locked to the R-peak, typically 200–600 ms after it. Being R-coherent, it has $\mathbb{E}[s_k(p)] \neq 0$ and therefore **survives averaging and lands in the template** — it does *not* decay with N_r . It is a bias, not leakage:

$$\mathbb{E}[\hat{d}_r(p)] = \underbrace{d_{\text{CFA}}(p)}_{\text{volume-conducted, want to remove}} + \underbrace{d_{\text{HEP}}(p)}_{\text{neural, R-locked, may NOT want to remove}}. \quad (7)$$

Consequence: synchronous averaging cannot distinguish CFA from HEP, because both are R-locked. If you subtract the template you remove the HEP too. Two levers separate them (neither is more averaging): *latency* — CFA is QRS-synchronous (near $p = 0$), HEP sits 200–600 ms later; and the *FIR residual* of the earlier note — the part of the R-locked template that a filtered ECG explains is volume-conducted CFA, the part it does not explain at HEP latency is the candidate HEP. This is the non-circular use of the FIR fit: after R-locking is confirmed, it splits the R-locked template into cardiac-field vs. cardiac-triggered-neural.

(b) Respiratory sinus arrhythmia (RSA) — a *variance*-inflation term

Respiration modulates the beat interval RR_k (your Assumption 2) and *also* modulates EEG — via arousal, and via respiration-driven electrode micro-motion and contact impedance. Because one breath spans several beats ($\sim 0.2\text{--}0.4\text{ Hz}$ vs. $\sim 1\text{ Hz}$), respiration-linked EEG is *slowly varying across beats*: it is neither cleanly R-locked (so it is not a fixed bias like the HEP) nor cleanly independent (so it does not average as $1/\sqrt{N_r}$). It appears as positive short-lag correlation c_{ij} between neighboring beats, which is exactly the term that drives $N_{\text{eff}} < N_r$ in Eq. (6). Net effect: RSA-coupled EEG **inflates the residual leakage above the ideal floor** rather than biasing the template shape. It is the physical reason your empirical leakage exceeds $1/\sqrt{N_r}$.

Summary of the three regimes.

Component	Relation to R-peak	Effect on template	Scaling
Free-running alpha, noise	asynchronous	averages out	$\text{SD}/\sqrt{N_r}$ (leakage)
Volume-conducted CFA	phase-coherent	retained (target)	bias, no decay
HEP (neural)	phase-coherent	retained (confound)	bias, no decay
RSA-coupled EEG	slow, partial coher.	excess leakage	$\text{SD}/\sqrt{N_{\text{eff}}}$, $N_{\text{eff}} < N_r$

5 What to put in the paper, and how to measure it

Text. Replace any “sampled at heart rate” wording with the asynchrony statement:

By Assumption 1, $s(t)$ and $n(t)$ are asynchronous with the R-peak: across beats the cardiac phase relative to ongoing EEG rhythms is uniformly distributed, so for each phase

index p the ensemble $\{s_k(p)\}$, $\{n_k(p)\}$ is zero-mean and (near-)incoherent. Its sample mean converges as $O(1/\sqrt{N_r})$ in standard deviation, while the R-locked disturbance $d(p)$ adds coherently. A small excess above the ideal floor reflects respiration-mediated (RSA) correlation between beats, i.e. an effective count $N_{\text{eff}} < N_r$; a residual R-locked neural component (HEP) survives as a bias and is treated in scope.

Measurements (one run on sub-001).

1. **Leakage floor / R-locking test.** Rebuild the template on phase-scrambled R-peaks (same N_r , shuffled intervals). The scrambled template's amplitude is your empirical leakage floor $\text{SD} / \sqrt{N_{\text{eff}}}$; the true-vs-scrambled ratio certifies the deflection is R-locked (CFA/HEP) rather than an averaging artifact. This directly *measures* N_{eff} .
2. **Alpha preservation.** Before/after 10 Hz spectral check — absence of a notch confirms free-running alpha was not removed (asynchronous $\Rightarrow$ untouched, as derived).
3. **CFA vs. HEP split.** FIR-explained fraction near $p = 0$ (CFA) vs. unexplained R-locked energy at 200–600 ms (HEP candidate), reported by latency.

Bottom line: $s(t)$ vanishes because it is phase-incoherent with the R-peak, and its finite-record residual falls as $\text{SD} / \sqrt{N_r}$ ($\text{SD} = \sqrt{\text{Var}}$; the radical is why it is $\sqrt{N_r}$). Heartbeat- and breath-driven *real* EEG breaks the ideal picture in two named ways: the HEP adds a non-decaying *bias* to the template, and RSA adds beat-to-beat correlation that *inflates the leakage* to $\text{SD} / \sqrt{N_{\text{eff}}}$ with $N_{\text{eff}} < N_r$.